



**BAHIR DAR UNIVERSITY**

**BAHIR DAR INSTITUTE OF TECHNOLOGY(BIT)**

**FACULTY OF COMPUTING**

**Operating System And System Programming**

**INDIVIDUAL PROJECT**

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# 1 .Introduction

Operating systems are the core software component of a computer system that manages hardware resources such as CPU, memory, storage, and input/output devices. They also provide a platform for running application software and ensure efficient communication between hardware and software components.

In traditional computing environments, installing and testing operating systems directly on physical machines can be risky because it may affect existing data, system configuration, or hardware stability. It is also time-consuming and limits flexibility when multiple operating systems are required for learning or testing purposes.

To overcome these limitations, virtualization technology is widely used in modern computing systems. Virtualization allows a single physical machine to run multiple virtual machines, each behaving like an independent computer system. This is achieved using virtualization software such as Oracle VM VirtualBox or VMware Workstation, which creates an isolated environment for operating system installation and execution.

In this assignment, Manjaro Linux is installed inside a virtual machine using VirtualBox on a Windows host system. Manjaro Linux is an Arch-based Linux distribution that is widely used for learning, development, and system programming due to its simplicity, performance, and strong community support.

The motivation of this practical work is to gain hands-on experience in operating system installation, understand the virtualization process, and explore system-level operations in a safe environment. It also helps in developing foundational skills in system programming, troubleshooting, and OS configuration without affecting the host operating system.

## 2.Objectives

The primary objective of this assignment is to develop a clear understanding of operating system installation using virtualization technology. It focuses on gaining both theoretical knowledge and practical experience in setting up an operating system within a virtual environment, ensuring that the host system remains unaffected during experimentation and learning.

Another key objective is to understand how virtualization tools such as Oracle VM VirtualBox or VMware Workstation function in creating and managing virtual machines. This includes learning how system resources such as CPU, memory (RAM), storage, and network interfaces are allocated to a virtual machine to simulate a real physical computer environment.

The assignment also aims to provide hands-on experience in installing and configuring Manjaro Linux inside a virtual machine. Through this process, the user becomes familiar with the installation workflow, including booting from ISO images, disk partitioning, user account creation, and system initialization.

In addition, this work is intended to improve understanding of virtualization in modern computing systems. Virtualization plays an important role in cloud computing, software development, system testing, and educational environments by allowing multiple operating systems to run simultaneously on a single physical machine efficiently and securely.

Another important objective is to introduce basic system-level concepts, including system configuration and preparation for system programming tasks. This helps build a foundation for working with Linux-based environments where system calls and low-level programming are commonly used.

Finally, the assignment aims to develop problem-solving and troubleshooting skills by dealing with common installation and configuration issues such as ISO mounting errors, boot failures, and system setup problems, which are essential skills in real-world system administration.

## 3.REQUIREMENTS

In order to successfully install and run an operating system in a virtual environment, it is necessary to ensure that both hardware and software requirements are properly met. These requirements ensure that the system runs smoothly, the virtual machine performs efficiently, and there are no performance or compatibility issues during installation and execution.

### 3.1 Hardware Requirements

**Processor (CPU):** A modern multi-core processor is required, preferably Intel or AMD processors that support hardware virtualization technology such as Intel VT-x or AMD-V. This feature is essential because it allows the virtual machine to run efficiently by directly utilizing CPU capabilities.

**RAM (Memory):** A minimum of 4GB RAM is required; however, 8GB or more is recommended for better performance. RAM is crucial because both the host operating system (Windows) and the virtual machine (Manjaro Linux) share memory resources.

**Storage:** At least 20GB to 50GB of free disk space is required. This space is used to store the virtual machine files, operating system installation files, and additional system data. Faster storage such as SSD is recommended for improved performance.

**BIOS/UEFI Support:** Virtualization technology must be enabled in the BIOS or UEFI settings of the system. Without enabling this feature, the virtual machine may not boot or perform properly.

**Graphics Support (Optional):** A basic graphics adapter is sufficient, but better GPU support may improve performance when running graphical applications inside the virtual machine

- **Virtualization Software:** Oracle VM VirtualBox or VMware Workstation is used to create and manage virtual machines.
- **Operating System ISO File:** The installation image of Manjaro Linux is required to install the operating system inside the virtual machine.
- **Host Operating System:** A working host operating system such as Windows is required to run the virtualization software.
- **Optional Tools:** Office suite or text editor for documentation and screenshot preparation.

**RAM (Memory):** A minimum of 4GB RAM is required; however, 8GB or more is recommended for better performance. RAM is crucial because both the host operating system (Windows) and the virtual machine (Manjaro Linux) share memory resources.

**Storage:** At least 20GB to 50GB of free disk space is required. This space is used to store the virtual machine files, operating system installation files, and additional system data. Faster storage such as SSD is recommended for improved performance.

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**Graphics Support (Optional):** A basic graphics adapter is sufficient, but better GPU support may improve performance when running graphical applications inside the virtual machine.

## 3.2 .Software Requirements

The software requirements include all programs and tools required to create, manage, and install the operating system inside the virtual environment.

- **Virtualization Software:** Oracle VM VirtualBox or VMware Workstation is required to create and manage virtual machines. These tools act as a bridge between the physical hardware and the virtual operating system, allowing multiple operating systems to run on a single device.
- **Operating System ISO File:** The installation file (ISO image) of Manjaro Linux is required. This ISO file contains the complete operating system setup and is used to install the OS inside the virtual machine.
- **Host Operating System:** A stable host operating system such as Windows is required to run the virtualization software. The host OS provides the environment in which virtual machines operate.
- **Device Drivers and Extensions:** VirtualBox Extension Pack or VMware Tools may be required to improve performance, enable better graphics support, and allow features like shared clipboard and file transfer between host and guest systems.

- **Internet Connection (Recommended):** An internet connection is useful during installation for downloading updates, packages, and additional software required for system configuration.
- **Basic Tools:** Tools such as a web browser, file manager, and text editor are required for downloading ISO files, managing virtual machine files, and preparing documentation.

# Installing Virtualization Software

## Purpose of Virtualization Software

Virtualization software enables a single physical computer to run multiple operating systems simultaneously by creating virtual machines (VMs). Each virtual machine operates as an independent system with its own virtual CPU, memory, storage, and network interface.

Tools such as Oracle VM VirtualBox and VMware Workstation are widely used virtualization platforms that simulate real hardware environments. This allows users to install and test different operating systems, such as Manjaro Linux, without affecting the main (host) operating system.

Virtualization is especially useful in system programming, software development, and cybersecurity because it provides a safe and controlled environment for experimentation, testing, and learning.

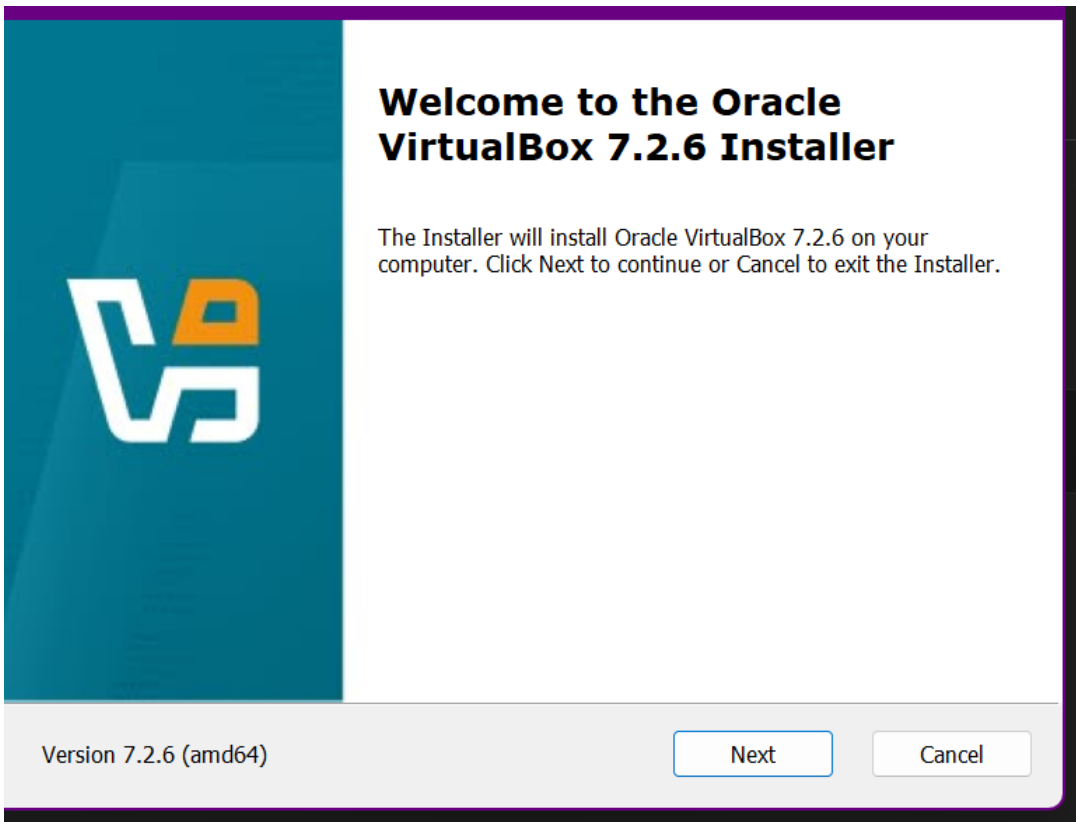
## Downloading the Software

The virtualization software was downloaded from its official website to ensure authenticity, security, and access to the latest stable version. Choosing the correct version compatible with the host operating system (e.g., Windows 10/11 64-bit) is important to avoid installation issues and ensure optimal performance.

During the download process, factors such as system architecture (32-bit or 64-bit), available system resources, and software requirements were considered. The latest version was selected because it includes performance improvements, bug fixes, and better hardware compatibility.

After downloading, the installer file (e.g., .exe for Windows) was saved in a known directory for easy access during installation.

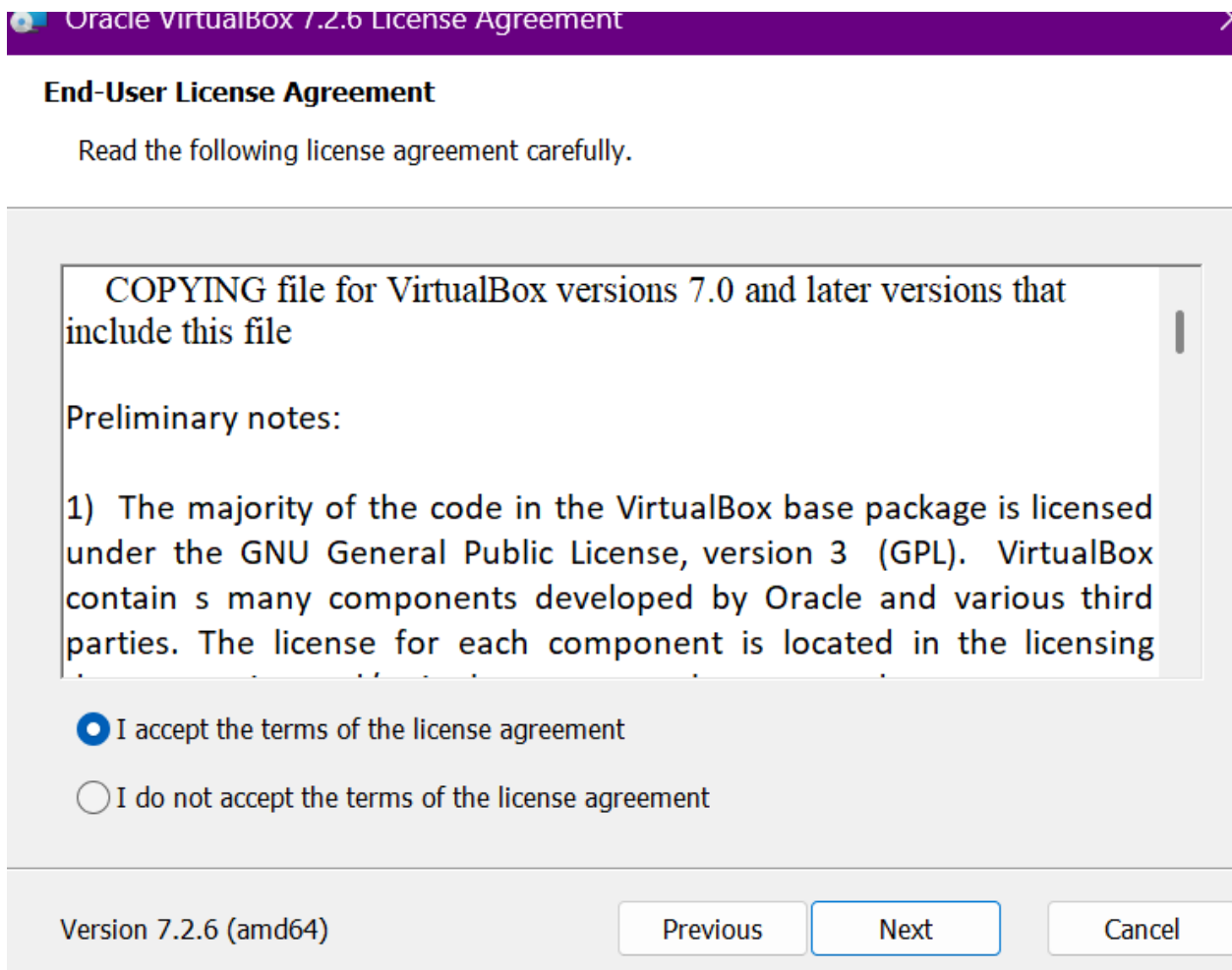




Link: <https://www.virtualbox.org/wiki/Downloads>

**Description:**

This link provides the official download page for Oracle VM VirtualBox. Download the version compatible with your operating system to ensure proper installation and performance.

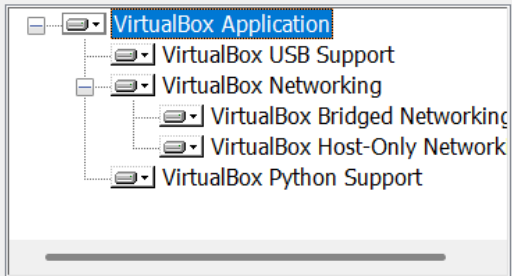


**Before initiating the installation process, the user must review and accept the predefined terms and conditions of the virtualization software.**

## Custom Setup

Select the features to install

Click on the icons in the tree below to change the way features will be installed.



Oracle VirtualBox 7.2.6 application.

This feature requires 233MB on your hard drive. It has 3 of 3 subfeatures selected. The subfeatures require 944KB on your hard drive.

Installation Location: C:\Program Files\Oracle\VirtualBox\

Change

Check Disk Space

Version 7.2.6 (amd64)

Previous

Next

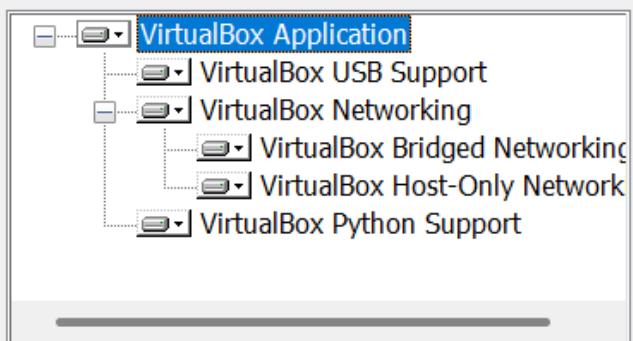
Cancel

Once the required information and settings are confirmed, the 'Next' button is selected to advance to the next stage of the installation process.

## Custom Setup

Select the features to install

Click on the icons in the tree below to change the way features will be installed.



Oracle VirtualBox 7.2.6 application.

This feature requires 233MB on your hard drive. It has 3 of 3 subfeatures selected. The subfeatures require 944KB on your hard drive.

Installation Location: C:\Program Files\Oracle\VirtualBox\

Change

Check Disk Space

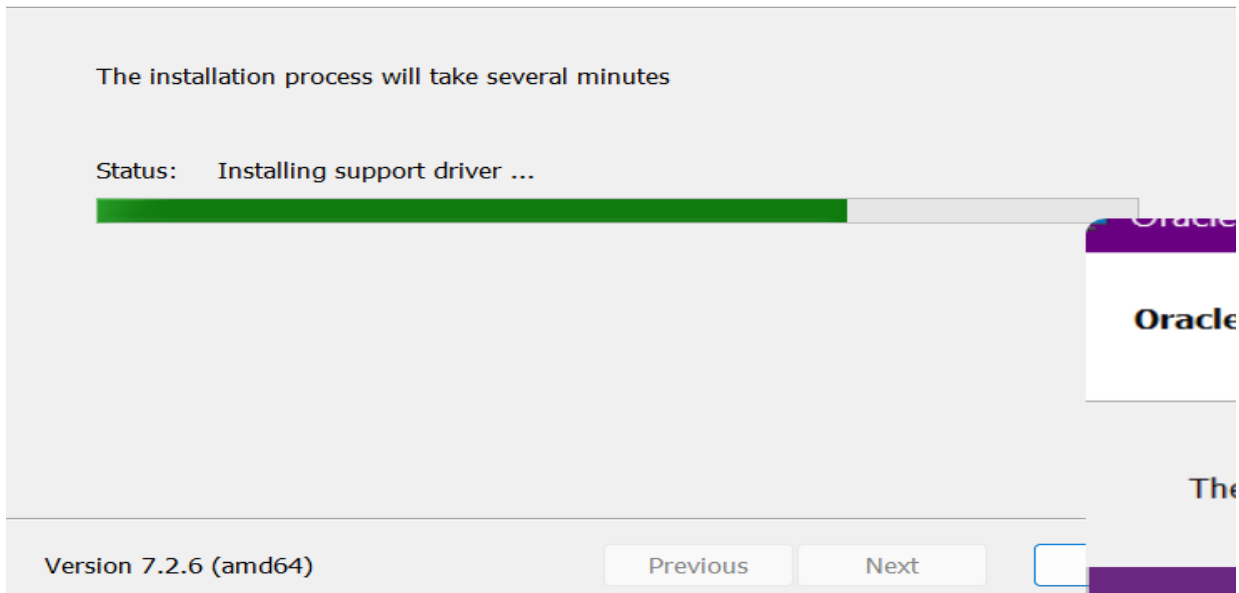
Version 7.2.6 (amd64)

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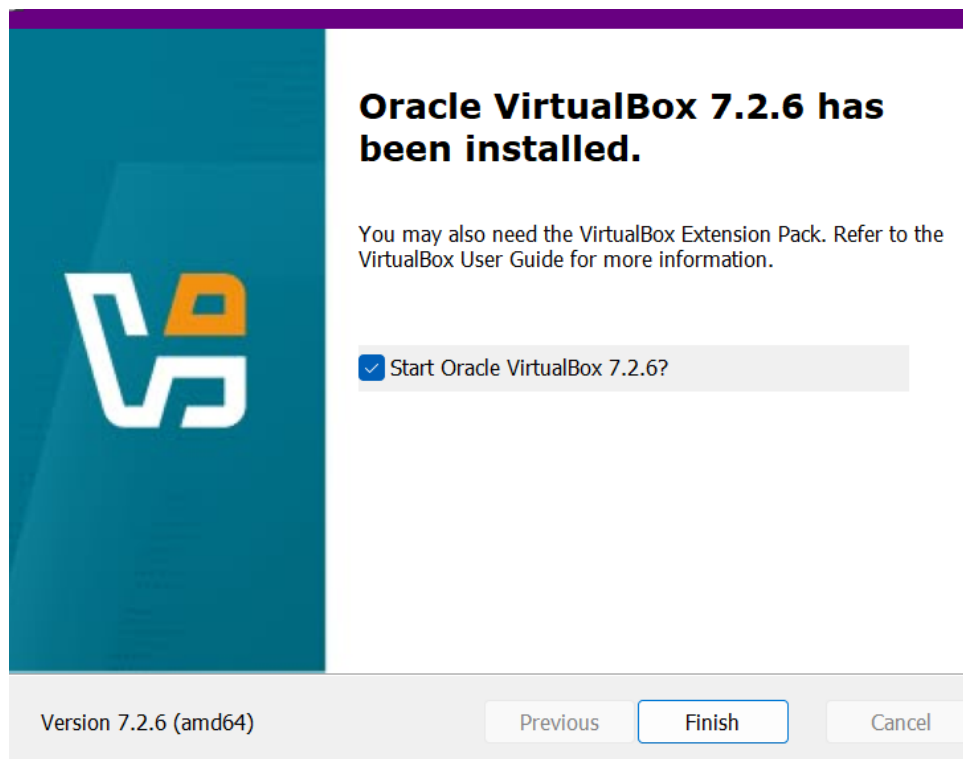
Next

Cancel

## Oracle VirtualBox 7.2.6



At this stage, all selected configurations and settings are reviewed to ensure accuracy before proceeding with the installation. The setup wizard displays a summary of the chosen options, including installation path and system configurations. Once verified, the installation process is initiated by selecting the 'Install' button. The system then begins copying and configuring the necessary files for successful installation.



After the installation process is completed, the setup wizard displays a confirmation message indicating successful installation. The 'Finish' button is then selected to close the installer, and the system may prompt for a restart to apply all configurations. Upon restarting, the machine boots into the newly installed Manjaro Linux operating system. This confirms that the installation was completed successfully and the system is ready for use.”

## What is Manjaro ISO?

The Manjaro Linux ISO is a disk image file that contains all the necessary files required to install the Manjaro operating system. An ISO file is essentially a digital copy of a physical installation disk (such as a DVD), packaged into a single file format that can be used to install or run an operating system.

The Manjaro ISO includes the complete operating system environment, including the Linux kernel, system libraries, desktop environment (such as XFCE, KDE, or GNOME), and essential applications. It is designed to be bootable, meaning it can be used to start a system directly and launch the installation process.

In a virtual environment, the ISO file is attached to a virtual machine and acts as a virtual installation disk. When the virtual machine is started, it boots from the ISO file, allowing the user to install Manjaro Linux step by step without needing any physical media.

Additionally, Manjaro ISO files often support live mode, which allows users to run the operating system temporarily without installing it. This feature is useful for testing hardware compatibility, exploring the interface, or troubleshooting before performing a full installation.

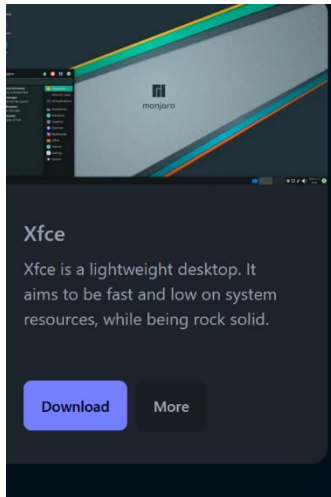
## **Downloading Manjaro ISO**

Before beginning the installation process, it is necessary to download the ISO file of Manjaro Linux. The ISO file contains all the essential components required to install the operating system, including system files, applications, and the installation environment. It serves as a bootable image that allows the virtual machine to start and launch the Manjaro installer.

The ISO file is obtained from the official Manjaro website to ensure security, reliability, and access to the latest stable version. Selecting the appropriate edition (such as XFCE, KDE, or GNOME) is important based on system performance and user preference. Once downloaded, the ISO file will be used in later steps to initiate the installation within the virtual environment.

## **Manjaro XFCE Edition**

The XFCE edition of Manjaro Linux is a lightweight and efficient desktop environment designed to provide a fast and user-friendly computing experience. It is known for its low system resource usage, stability, and simplicity, making it suitable for both beginners and systems with limited hardware capabilities. The XFCE version was selected for this project because virtualization environments typically have limited allocated resources such as RAM and CPU. Using XFCE ensures smooth performance of the virtual machine while maintaining essential features and usability. Additionally, its clean interface and reliability make it ideal for learning and installing an operating system in a virtual environment.

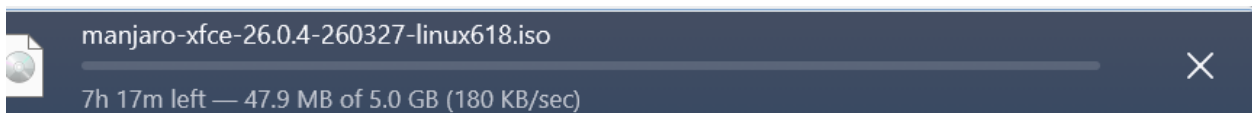


## Completion of Manjaro XFCE ISO Download

At this stage, the download of the XFCE edition of Manjaro Linux has been successfully completed. The ISO file, which contains all necessary installation components, is now stored on the local system and ready for use in the virtualization process.

The downloaded file will be utilized in the subsequent steps by attaching it to the virtual machine as a bootable installation medium. This ensures that the virtual machine can initiate the Manjaro installation environment.

With the ISO file prepared and available, the system is now ready to proceed to the next phase, which involves creating and configuring the virtual machine and attaching the ISO file for installation.



## Create Virtual Machine

Link: <https://www.virtualbox.org/manual/ch01.html#create-vm>

This guide demonstrates how to create a new virtual machine using Oracle VM VirtualBox. During this process, essential configurations such as the virtual machine name, operating system type (Linux/Arch), memory (RAM), and storage capacity are defined.



These settings are important because they determine how the virtual machine will perform and interact with the host system. Proper configuration ensures that the virtual machine can efficiently run the selected operating system, such as Manjaro Linux, without performance issues.

## Create Virtual Machine

The virtual machine is created using the installed virtualization software.

- Open the virtualization software
- Click “New” to create a new virtual machine
- Name: Enter your full name (important for identification)
- Type: Linux
- Version: Arch Linux / Other Linux (64-bit)
- Select appropriate settings based on system compatibility

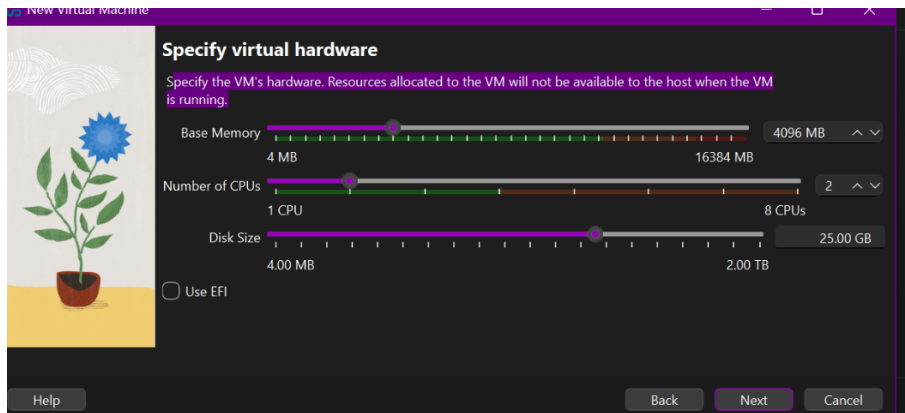
This step defines the basic identity and operating system type of the virtual machine.

## Allocate System Resources

System resources are assigned to ensure smooth performance of the virtual machine.

- RAM: 2 GB – 4 GB (depending on host system capacity)
- CPU: 2 processor cores recommended
- Disk Space: 20 GB – 30 GB virtual hard disk

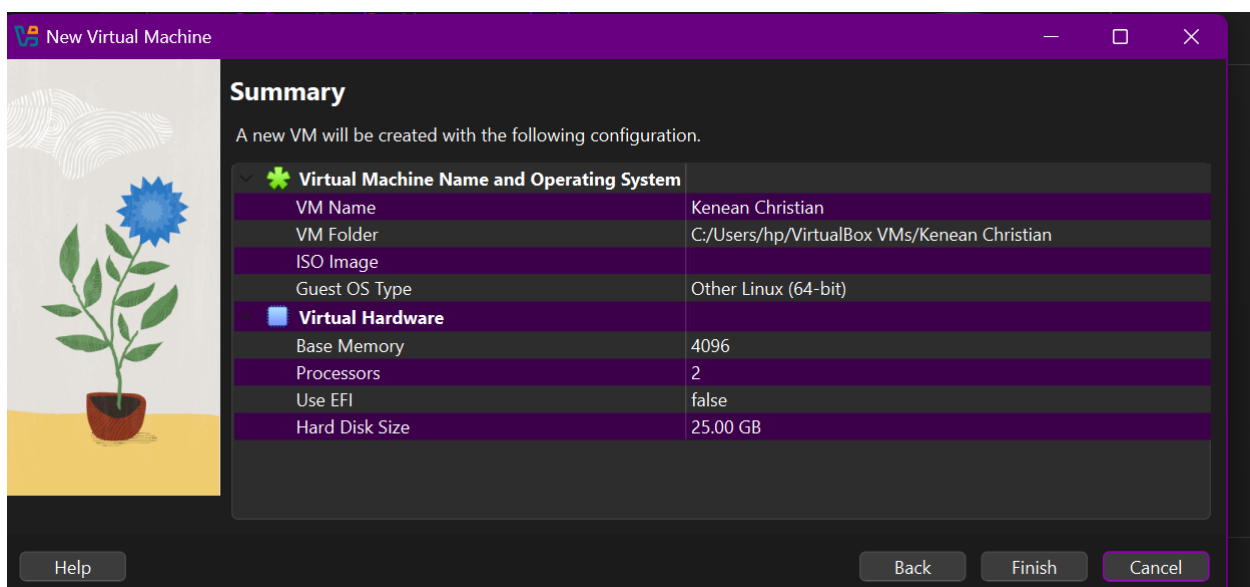
Proper allocation of resources is important to ensure that the virtual machine runs efficiently while maintaining stability on the host system. This configuration supports smooth operation of operating systems such as Manjaro Linux.

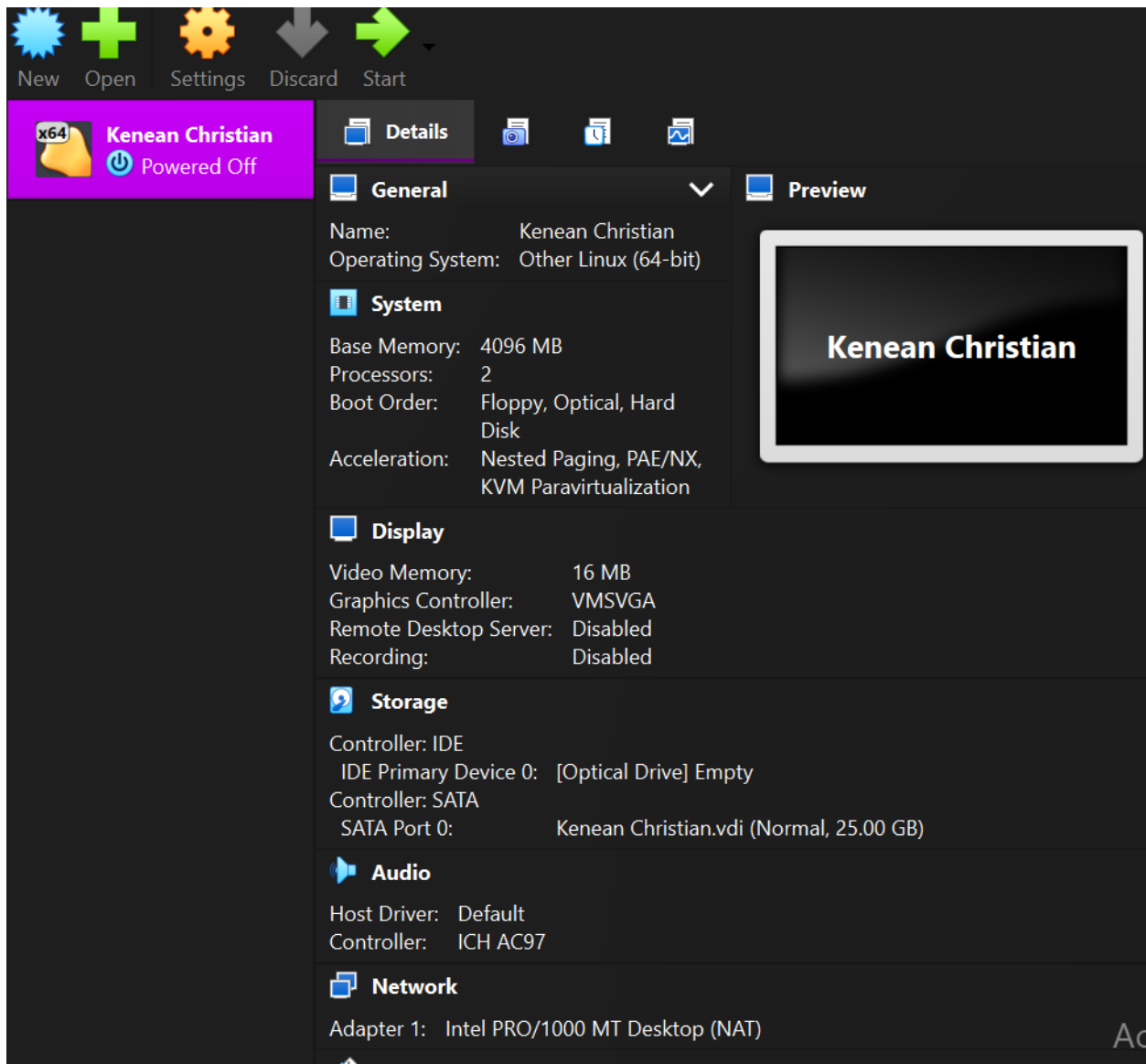


In this step, appropriate system resources were assigned to the virtual machine to ensure optimal performance during operation. The allocated resources included 2–4 GB of RAM, 2 CPU cores, and a virtual hard disk space ranging from 20–30 GB.

During the configuration process, the virtual machine was also named using the user’s full name, Kenean Christian, to ensure proper identification and organization within the virtualization environment.

These settings are important because they determine how efficiently the virtual machine runs on the host system. Proper allocation ensures that the virtual environment can smoothly support the installation and execution of operating systems such as Manjaro Linux without causing significant performance issues on the host computer.





## Attach Manjaro ISO File

### What is an ISO File?

An ISO file is a complete image of an operating system installation disk. It contains all the necessary files required to install the operating system. In this case, the ISO file of Manjaro Linux is used to install the OS on the virtual machine.

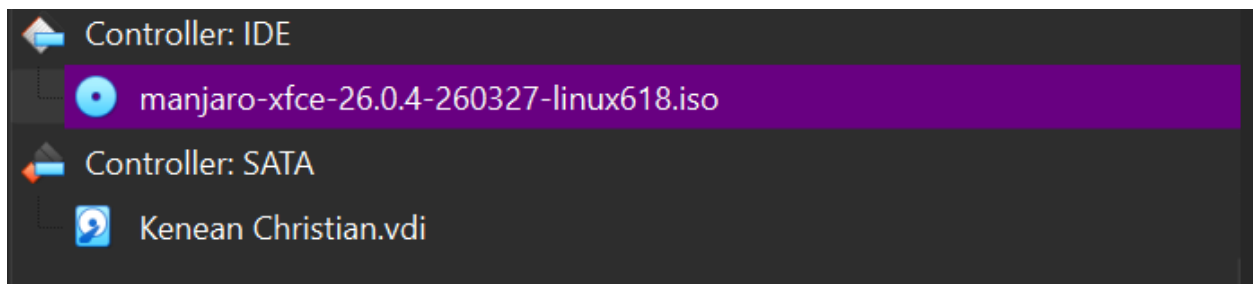
### Accessing Virtual Machine Settings

To attach the ISO file, the settings of the virtual machine are opened. Inside the settings menu, the **Storage** section is selected, where virtual drives such as hard disks and optical drives are managed.

### Attaching the ISO File

Under the storage controller (IDE or SATA), the empty optical drive is selected. Then, the Manjaro ISO file is chosen from the local system and attached to the virtual machine.

This step ensures that the virtual machine has access to the installation files needed to start the OS setup.

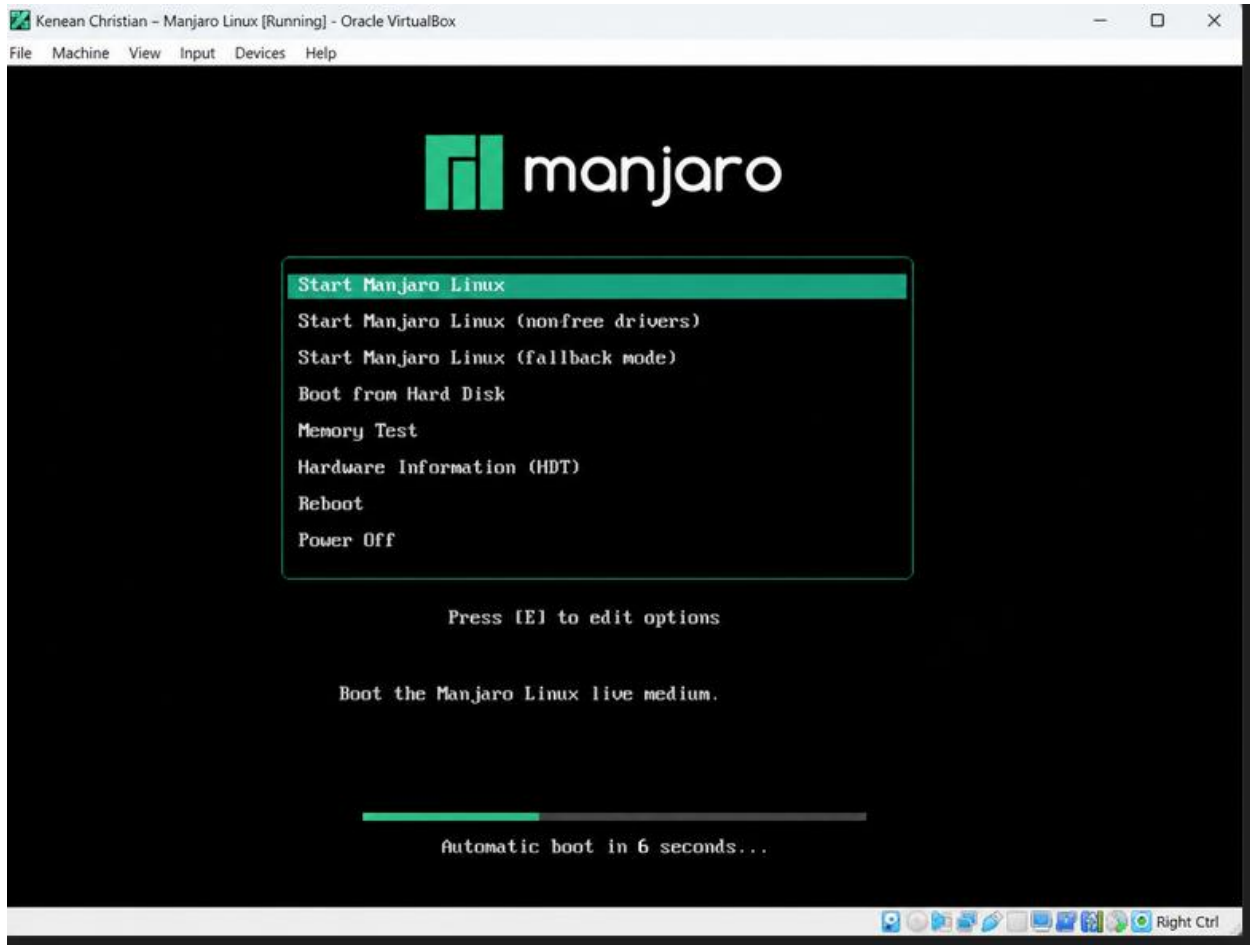


### Booting and Operating System Installation Process

After successfully attaching the ISO file of Manjaro Linux to the virtual machine, the system was started to initiate the booting process. The virtual machine was configured to boot from the mounted ISO file, allowing it to load the Manjaro live environment.

#### 5.1 Booting the Virtual Machine

When the virtual machine is powered on, it reads the ISO file and loads the Manjaro boot menu. From this menu, the option to start the live environment with open-source drivers was selected. This step is important because it prepares the system for installation without modifying the virtual hard disk immediately

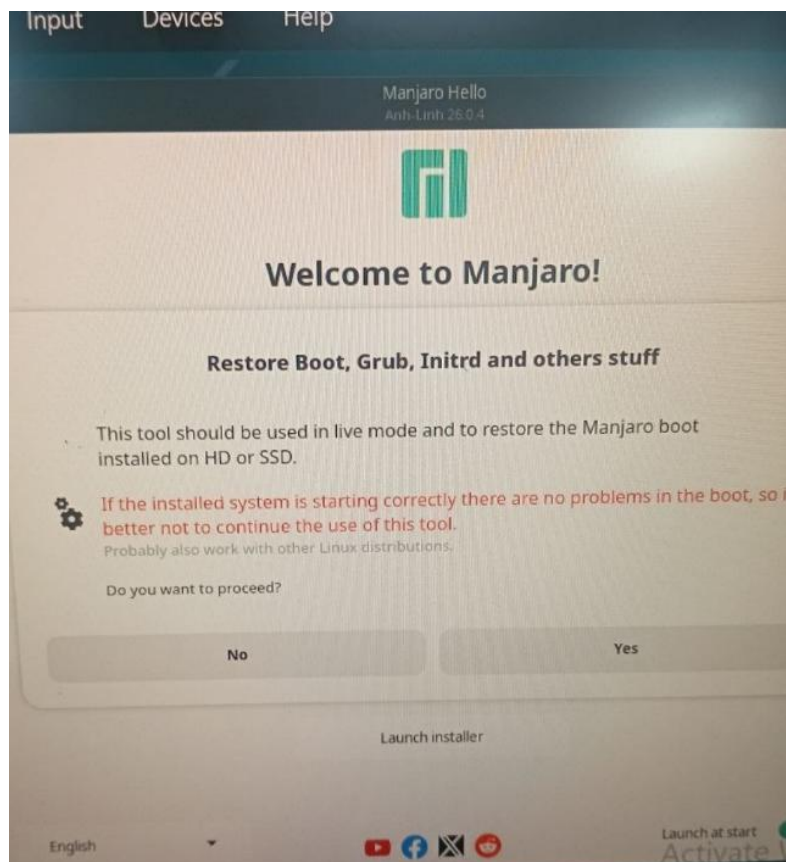


## Starting the Installer

At this stage, the installation process of Manjaro Linux is initiated from within the live desktop environment. After successfully booting into the system, the user opens the “Install Manjaro” application available on the desktop interface. This action launches the graphical installation wizard, which is responsible for guiding the user through the complete operating system setup process.

The installer provides a structured and user-friendly interface where all essential installation configurations will be completed in subsequent steps, such as language selection, region settings, disk partitioning, and user account creation. This step is important because it marks

the transition from the live environment into the actual installation phase on the virtual hard disk.

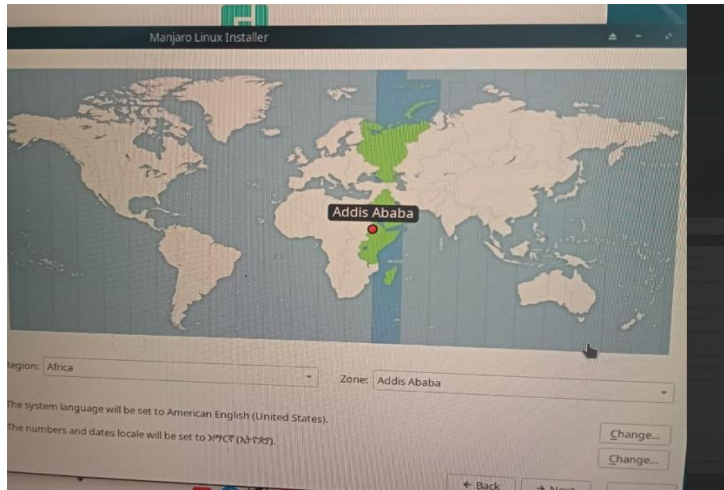


## **Language, Region, and Keyboard Setup**

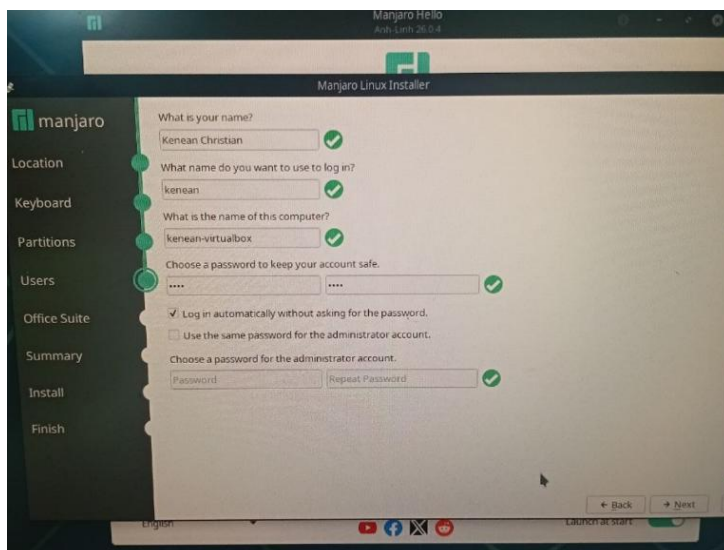
At this stage of the installation process of Manjaro Linux, the installer prompts the user to configure basic system settings such as language, region, and keyboard layout. These settings are essential as they determine how the operating system will communicate with the user and how input is interpreted.

The user selects the appropriate language for system display, sets the correct geographical region to ensure accurate time zone and system formatting, and chooses a suitable keyboard layout for proper input functionality. These configurations ensure that the operating system is personalized according to the user's location and preferences.

This step is important because incorrect settings at this stage may affect system usability, including time display, language interface, and keyboard input accuracy.

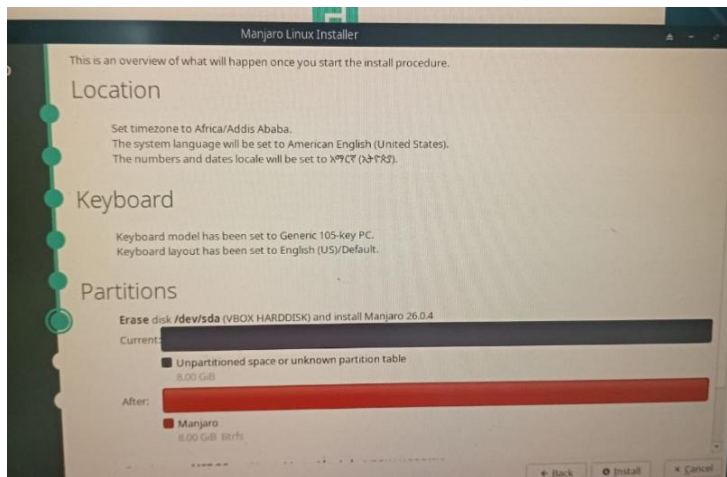


The region selection determines the geographical location of the system, which is used to configure the correct time zone, date format, and system language preferences. In the installation of Manjaro Linux, the appropriate country or region was selected based on the user's actual location. This ensures that the system time is accurate and synchronized with local standards. Proper region configuration is essential for correct system behavior and localization settings.





During the installation of Manjaro Linux, the user account setup stage requires the creation of personal login credentials. In this step, the full name **Kenean Christian** was entered to identify the system user, followed by the creation of a username used for system login. A secure password was then set to protect the account and prevent unauthorized access. These credentials are essential for authentication, allowing the user to securely access and manage the operating system after installation. This step also helps personalize the system environment for individual use.



## Keyboard Layout Selection

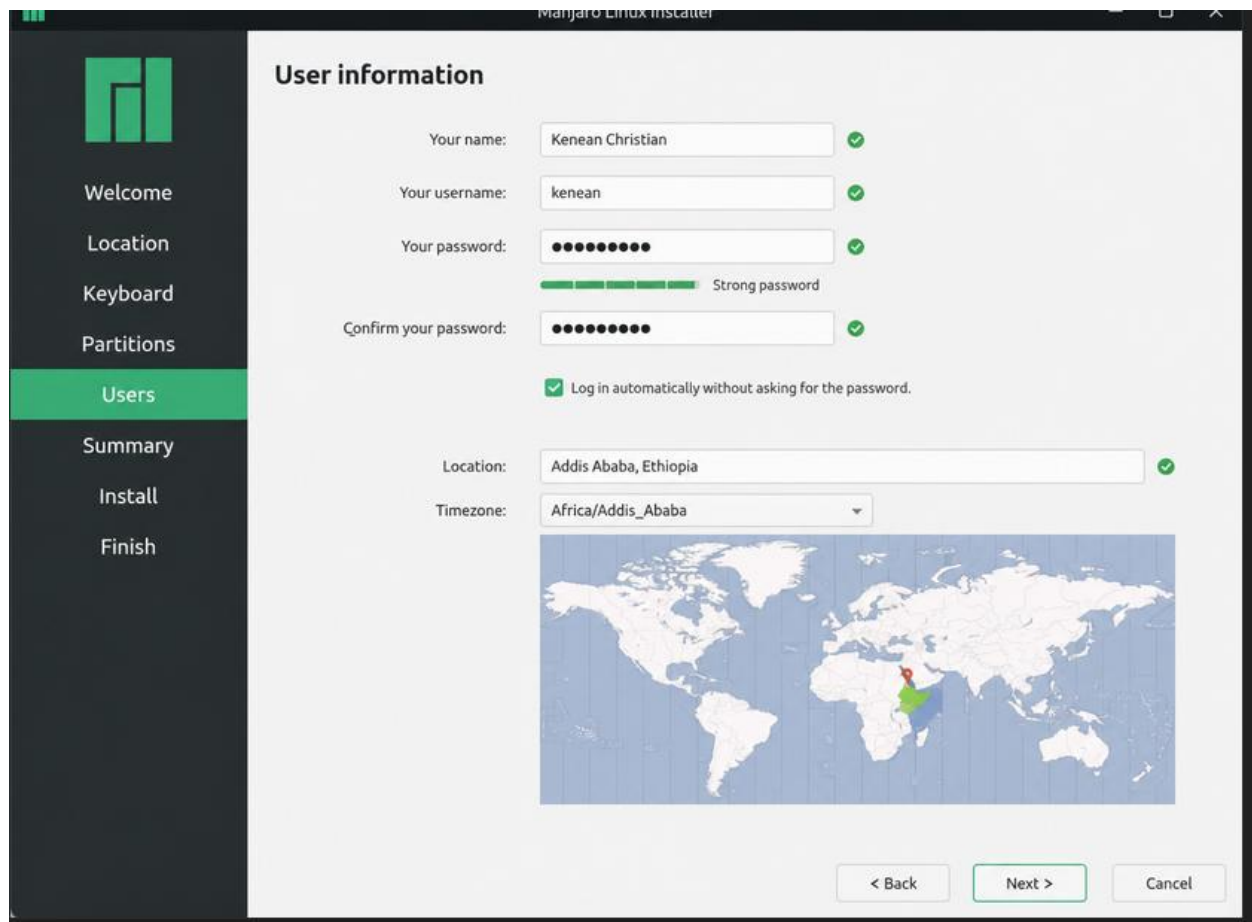
During the installation of Manjaro Linux, the keyboard layout is configured to ensure that all input from the keyboard is correctly interpreted by the system. The appropriate keyboard layout was selected based on the user's preference and physical keyboard type. This step is important because incorrect keyboard settings may lead to input errors, such as wrong characters appearing when typing. Proper configuration ensures accurate and efficient text input throughout the system.

## User Account and Location Configuration

At this stage of the installation process of Manjaro Linux, the user account details and system location are configured. The setup interface requires the entry of personal information such as full name, username, and password to create a secure login account for system access. In this case, the full name **Kenan Christian** was used to identify the system user.

Additionally, the location is set to **Addis Ababa, Ethiopia**, which ensures that the system automatically configures the correct time zone, regional settings, and date formats. This geographical configuration is important for system synchronization and proper localization of applications and services.

A secure password is also created to protect the user account and prevent unauthorized access to the system. This step ensures that the operating system is personalized, secure, and correctly configured for the user's environment.



The image shows the 'User information' screen of the Manjaro Linux installer. On the left is a dark sidebar with a green logo and a list of steps: Welcome, Location, Keyboard, Partitions, Users (highlighted in green), Summary, Install, and Finish. The main area is light gray and contains the following fields and options:

- Your name:** Kenean Christian (with a green checkmark)
- Your username:** kenean (with a green checkmark)
- Your password:** [masked with dots] (with a green checkmark and a 'Strong password' indicator)
- Confirm your password:** [masked with dots] (with a green checkmark)
- ☒ Log in automatically without asking for the password.
- Location:** Addis Ababa, Ethiopia (with a green checkmark)
- Timezone:** Africa/Addis\_Ababa (dropdown menu)

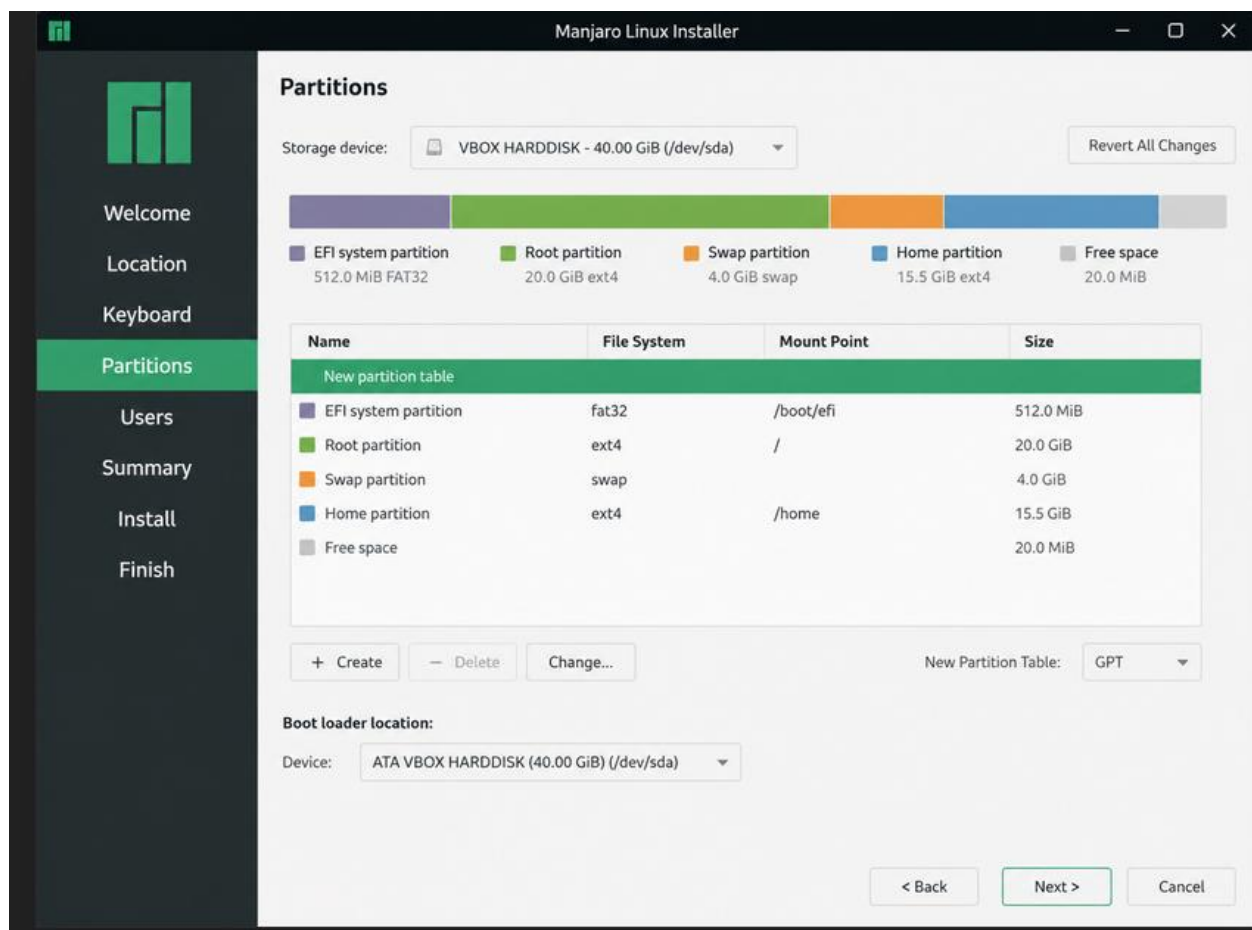
Below the timezone dropdown is a world map with a red pin in Ethiopia. At the bottom right are three buttons: '< Back', 'Next >' (highlighted in green), and 'Cancel'.

## Disk Partitioning

At this stage of the installation process of Manjaro Linux, the virtual hard disk is prepared and divided into different partitions to organize system storage effectively. The partitioning screen displays how the disk space is allocated for different purposes, including the root partition, swap space, and home directory.

The root partition (/) is used for installing the core operating system files, while the swap partition acts as virtual memory to support system performance when physical RAM is fully utilized. The home partition (/home) is used to store user files and personal data separately from system files, improving data management and system stability.

In this step, the appropriate disk layout is selected to ensure proper installation and efficient system performance within the virtual environment

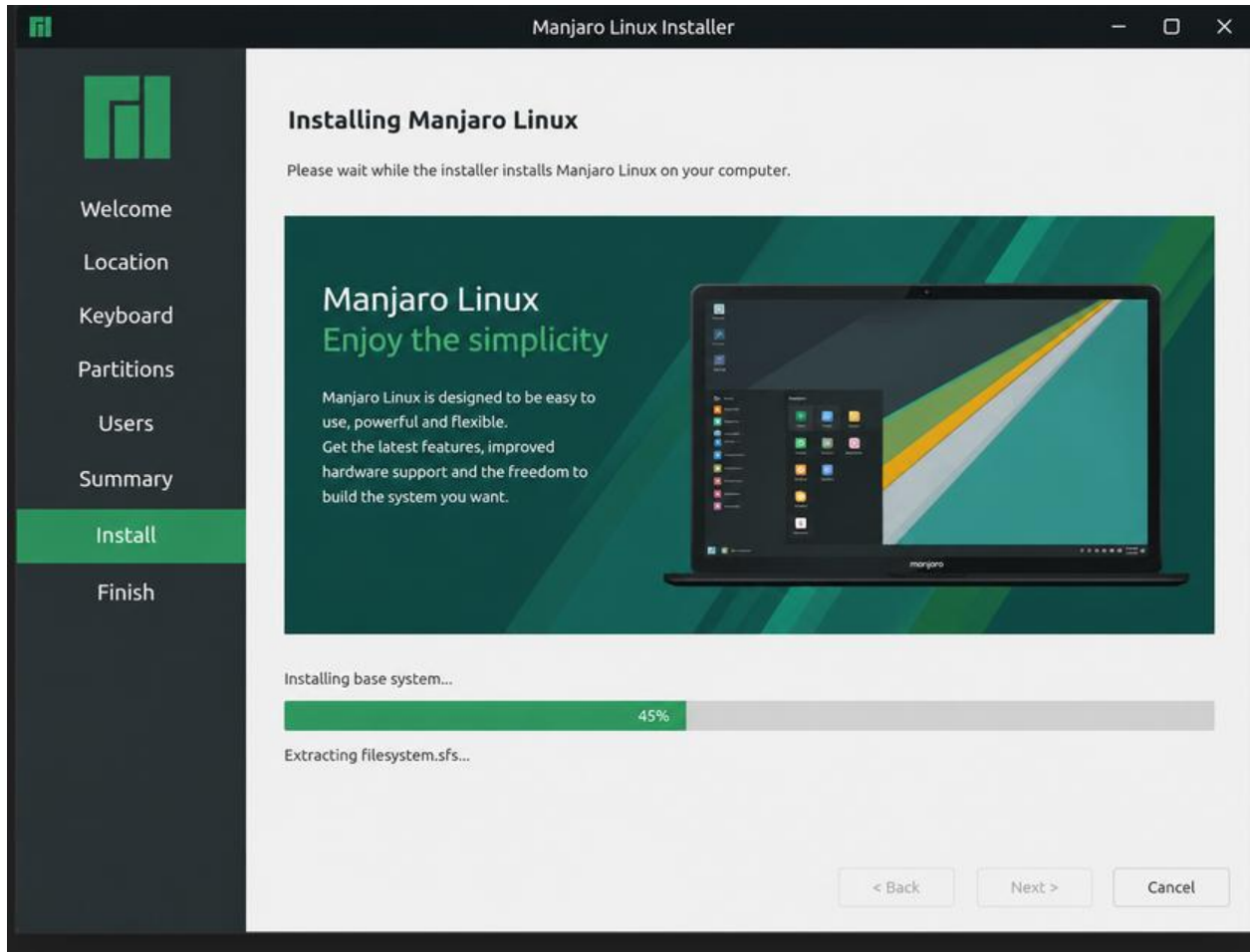


# System Installation Progress

At this stage of the installation process of Manjaro Linux, the operating system files are actively being installed onto the virtual hard disk. The installer begins copying essential system components, extracting packages, and configuring the base system required for proper operation.

A progress bar is displayed to indicate the current status of installation, showing the percentage of completion. In this phase, system files are being transferred and organized, including the filesystem setup and core system configuration. This process ensures that all necessary components are correctly installed for the operating system to function properly.

The installation process may take several minutes depending on system performance and allocated resources within the virtual machine. During this time, no user interaction is required until the process is completed.

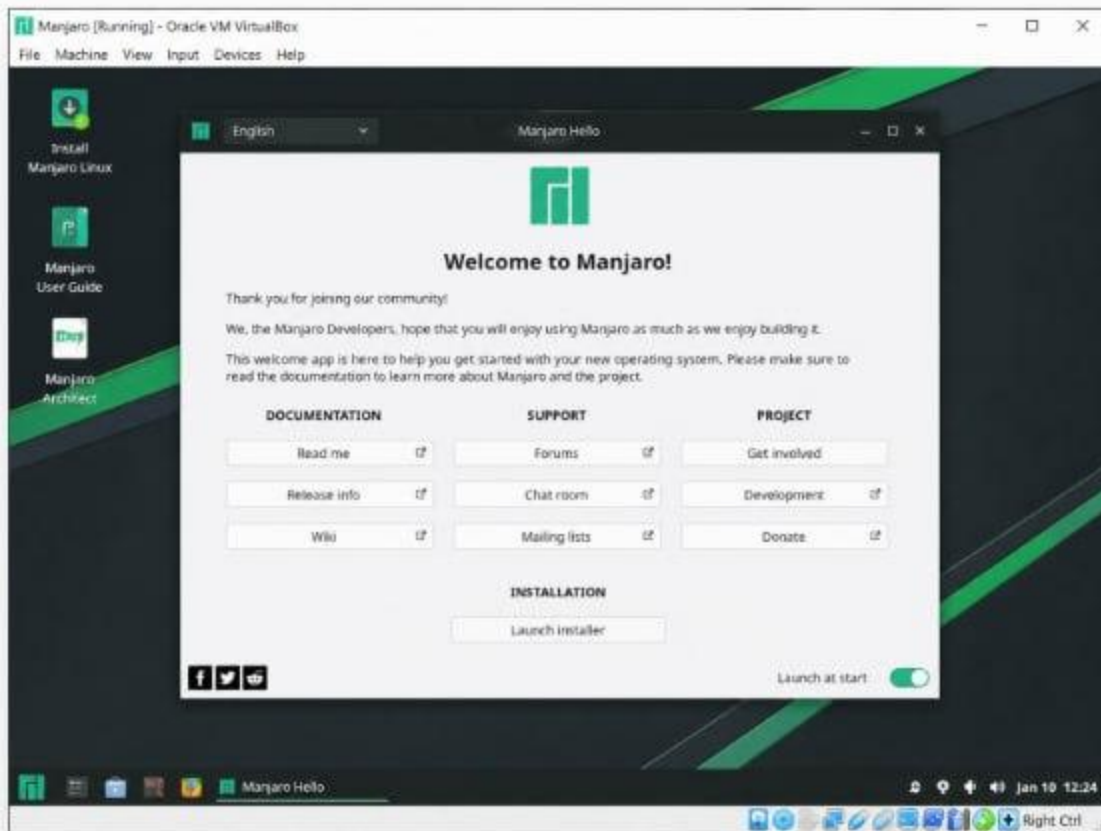


## Installation Completion and System Restart

At this final stage of the installation process of Manjaro Linux, the system confirms that the operating system has been successfully installed on the virtual machine. A completion message is displayed indicating that all installation tasks, including file copying, system configuration, and bootloader setup, have been successfully finalized.

The installer provides the option to either continue using the live environment or restart the system. In this step, the “Restart Now” option is selected to reboot the virtual machine. Before restarting, the installation media (ISO file) is removed to ensure that the system boots directly from the newly installed operating system.

After rebooting, the system loads into the installed Manjaro desktop environment, confirming that the installation process has been completed successfully and the system is ready for use.

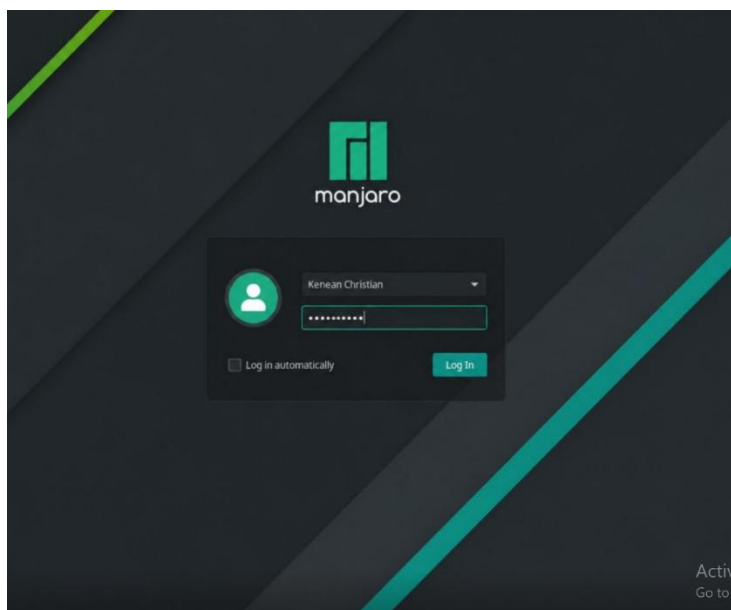


## Login Screen of Manjaro Linux

The login screen of Manjaro Linux is displayed after the system successfully boots from the installed virtual disk. This interface, managed by the system's display manager, serves as a security checkpoint before granting access to the desktop environment. At this stage, the user is required to enter the previously created username and password.

The login screen typically includes a user selection panel, a password input field, and system options such as shutdown or restart. It ensures that only authorized users can access the system and protects personal data and system configurations from unauthorized use.

Once the correct credentials are entered, the system authenticates the user and loads the XFCE desktop environment. This marks the transition from the system startup phase to full user interaction, where applications can be launched and system operations can be performed.





## Manjaro OS Desktop Ready

After completing the installation and restarting the system, the Manjaro Linux operating system successfully boots into the desktop environment (XFCE). The login process is completed using the previously created user credentials, after which the system loads the graphical user interface.

At this stage, the desktop environment is fully functional, indicating that the installation was successful. The user can now access system tools, open applications, and perform development tasks such as writing and executing programs. This confirms that the system is ready for practical usage, including compiling and running C programs related to system calls like `listen()`.



## System Call: listen()

The listen() system call is used in network programming to mark a socket as a **passive socket**, meaning it is ready to accept incoming connection requests from client applications. It is an essential function in the server-side implementation of the client-server architecture.

When a socket is created using the socket() system call and bound to a specific IP address and port using bind(), the listen() function is then called to transition the socket into a listening state. In this state, the operating system prepares a queue to store incoming connection requests until they are accepted by the server using the accept() system call.

The listen() function also defines the maximum number of pending connections that can be held in the queue. This helps manage multiple client requests efficiently and prevents overload on the server.

In summary, listen() plays a critical role in enabling server processes to wait for and manage incoming client connections in a controlled and organized manner.

## Working Concept

The working concept of the lseek() system call is based on adjusting the file pointer within an open file to control where the next read or write operation will occur. When SEEK\_SET is used, the file pointer moves relative to the beginning of the file by the specified offset. When SEEK\_CUR is used, the movement is based on the current position of the file pointer, allowing it to move forward or backward. When SEEK\_END is used, the pointer moves relative to the end of the file, which is useful for appending data or accessing content near the file's end.

This mechanism provides efficient random access to file content, making it an important feature in system-level file operations.

### ❖ Example Program using lseek()

A C program was written and executed in Manjaro Linux to demonstrate the working of the lseek() system call. The program opens a text file and uses lseek() with different reference values such as SEEK\_SET, SEEK\_CUR, and SEEK\_END. After adjusting the file pointer to various positions, the program reads selected portions of the file and displays the output on the screen.

This demonstrates how lseek() enables flexible navigation within a file and allows efficient access to specific data without reading the entire file sequentially.

```

1  /* Sample program to demonstrate listen() system call in C */
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <sys/socket.h>
5  #include <netinet/in.h>
6  #include <unistd.h>
7
8  int main() {
9      int server_fd, backlog = 5;
10     struct sockaddr_in address;
11
12     // 1. Create socket
13     server_fd = socket(AF_INET, SOCK_STREAM, 0);
14     if (server_fd == 0) {
15         perror("socket failed");
16         exit(EXIT_FAILURE);
17     }
18
19     // 2. Set address
20     address.sin_family = AF_INET;
21     address.sin_addr.s_addr = INADDR_ANY;
22     address.sin_port = htons(8080);
23
24     // 3. Bind socket
25     if (bind(server_fd, (struct sockaddr *)&address, sizeof(address)) < 0) {
26         perror("bind failed");
27         exit(EXIT_FAILURE);
28     }
29
30     // 4. Listen for incoming connections
31     if (listen(server_fd, backlog) < 0) {
32         perror("listen failed");
33         exit(EXIT_FAILURE);
34     }
35     printf("Server is listening on port 8080...\n");
36
37     return 0;

```

# Explanation of Socket Programming Library Functions

## 1. socket()

The `socket()` function is used to create a new communication endpoint. It defines the type of communication (such as TCP or UDP) and returns a socket descriptor, which is used for all further network operations. It is the starting point of any client-server communication.

## 2. bind()

The `bind()` function assigns a specific IP address and port number to a socket. This allows the system to know where the server application will listen for incoming connections. Without `bind()`, the socket cannot be linked to a network location.

## 3. listen()

The `listen()` function is used to mark a socket as a passive socket that can accept incoming connection requests. It also creates a queue to hold pending client connections until they are processed. This function is mainly used on the server side.

## 4. accept()

The `accept()` function is used to accept an incoming connection request from a client. Once a connection is accepted, it creates a new socket specifically for communication between the server and that client.

## 5. `connect()`

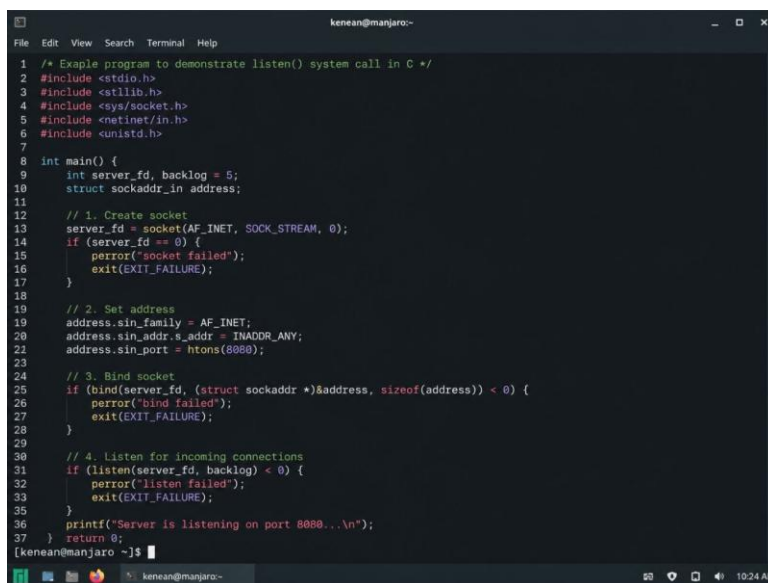
The `connect()` function is used by the client to establish a connection with a server. It sends a request to the server's IP address and port, initiating communication.

## 6. `read()` / `recv()`

These functions are used to receive data from the socket. They allow the program to read incoming messages sent by the other side of the connection.

## 7. `write()` / `send()`

These functions are used to send data through the socket. They transmit messages from the client or server to the connected peer.



```
1 /* Example program to demonstrate listen() system call in C */
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <sys/socket.h>
5 #include <netinet/in.h>
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34    }
35
36    printf("Server is listening on port 8080...\n");
37    return 0;
38 }
```

## Issues (Problems Faced)

During the installation of Manjaro Linux in the virtual environment, several challenges were encountered that affected the smooth progress of the setup process.

One of the main issues was related to **mouse integration within the virtual machine**. The mouse pointer was not properly synchronized with the virtual environment, making it difficult to navigate through the installation interface. This caused delays while selecting options and interacting with the installer.

Another issue occurred during the **ISO attachment and boot process**. Initially, the system did not boot correctly from the ISO file due to incorrect configuration in the virtual machine settings. As a result, the installation environment failed to load properly until the issue was identified.

In addition, **performance limitations** were observed during installation. The virtual machine operated slowly, especially when loading the live environment and running the installer. This was mainly due to limited allocation of system resources such as RAM and CPU.

There were also **initial configuration difficulties** when setting up the virtual machine. Selecting the correct operating system type, allocating appropriate disk space, and configuring hardware settings required careful attention, which made the early setup phase slightly challenging.

## Solutions

To resolve the issues encountered during the installation of Manjaro Linux, several corrective measures were applied.

First, the **mouse integration problem** was addressed by adjusting the virtual machine settings and enabling proper input capture. Additionally, installing or enabling virtualization tools (such as Guest Additions) helped improve mouse responsiveness and control.

Next, the **boot issue was resolved** by correctly attaching the Manjaro ISO file in the storage settings and ensuring that the virtual machine boot order prioritized the optical drive. This allowed the system to load the installation environment successfully.

To improve system performance, the **allocated resources were increased**, including assigning 2–4 GB of RAM and multiple CPU cores. This significantly enhanced the speed and responsiveness of the virtual machine during installation.

Furthermore, unnecessary applications running on the host system were closed to free up system resources. This helped reduce lag and ensured smoother execution of the installation process.

Finally, the installation process was restarted and followed carefully step by step. Adequate time was allowed for each stage, especially during system loading and installation phases, which ensured that the process completed successfully without interruption.

## Filesystem Support

Manjaro Linux primarily uses the **ext4 filesystem**, which is one of the most reliable and widely used filesystems in Linux-based operating systems. It is chosen because of its stability, performance, and ability to efficiently manage system data.

The ext4 filesystem organizes how files and directories are stored on the disk in a structured and efficient manner. It supports fast read and write operations, which improves overall system responsiveness and allows applications to load quickly. This makes the operating system perform smoothly even under moderate workloads.

Another important feature of ext4 is its support for large file sizes and large storage capacities. It can handle modern storage requirements effectively, making it suitable for both small and large-scale systems.

In addition, ext4 includes journaling capabilities, which help protect data by reducing the risk of corruption in case of unexpected shutdowns or system failures. Due to these advantages, ext4 is considered a secure and dependable filesystem for Manjaro Linux.

## Advantages and Disadvantages of Manjaro Linux

### Advantages

One of the main advantages of Manjaro Linux is its **user-friendly design**, especially compared to other Arch-based distributions. It provides an easy installation process and a clean interface, making it suitable for both beginners and advanced users.

Another major advantage is its **performance and efficiency**. The XFCE edition is lightweight and consumes fewer system resources, allowing it to run smoothly even in virtual machines or systems with limited hardware.

Manjaro also offers **rolling release updates**, meaning users receive the latest software updates continuously without needing to reinstall the operating system. This ensures access to up-to-date features and security improvements.

Additionally, it has **strong hardware support and package management**, making it easy to install and manage applications using built-in tools.

## Disadvantages

One disadvantage of Manjaro Linux is that, as a rolling release system, updates may sometimes introduce instability if not managed carefully.

Another limitation is that some **advanced configurations may require technical knowledge**, especially when dealing with system-level settings or troubleshooting.

In addition, although Manjaro has a good community, it is still **less widely supported than distributions like Ubuntu**, which may make finding solutions slightly more difficult in some cases.

## Conclusion

Manjaro Linux is a powerful and flexible Linux-based operating system that provides a balance between performance, usability, and modern features. Installing it using virtualization software such as Oracle VM VirtualBox demonstrates how operating systems can be deployed and tested in a safe and controlled environment.

The installation process involved several key steps, including creating a virtual machine, allocating system resources, attaching the ISO file, and completing the guided installation process. Each step plays an important role in ensuring that the system is correctly configured and fully functional.

Virtualization makes it possible to run multiple operating systems on a single physical machine without affecting the host system. This provides a practical environment for learning, experimentation, and development.

Overall, this project helped in understanding both operating system installation and virtualization concepts. It also provided hands-on experience in configuring virtual machines and working with Linux-based systems.

## Future Recommendation

For improved performance, it is recommended to allocate **at least 3GB–4GB of RAM** to the virtual machine instead of the minimum configuration. This will enhance system speed and reduce lag during both installation and regular use.

It is also beneficial to explore other operating systems such as Ubuntu, Kali Linux, and Windows Server to gain a broader understanding of different system environments.

In addition, practicing Linux command-line operations is highly recommended, as they are essential for efficient system management. Learning advanced topics such as system programming and networking will further strengthen technical skills and improve understanding of real-world computing systems.

## Virtualization in Modern Operating Systems

### What is Virtualization

Virtualization is a technology that allows a single physical computer to run multiple virtual machines, each acting as an independent system. These virtual machines can run different operating systems simultaneously while sharing the same hardware resources.

### Why Virtualization is Used

Virtualization improves **resource utilization**, reduces hardware costs, and provides a safe environment for testing and development. It allows users to experiment with different operating systems without affecting the main system.

It also enhances **security and isolation**, as each virtual machine operates independently. Problems in one system do not affect others.

### How Virtualization Works

Virtualization works through a software layer called a **hypervisor**, such as Oracle VM VirtualBox or VMware Workstation. The hypervisor manages system resources like CPU, memory, and storage, and distributes them among virtual machines.

Each virtual machine operates like a real computer, running its own operating system and applications independently while sharing the same physical hardware.



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